



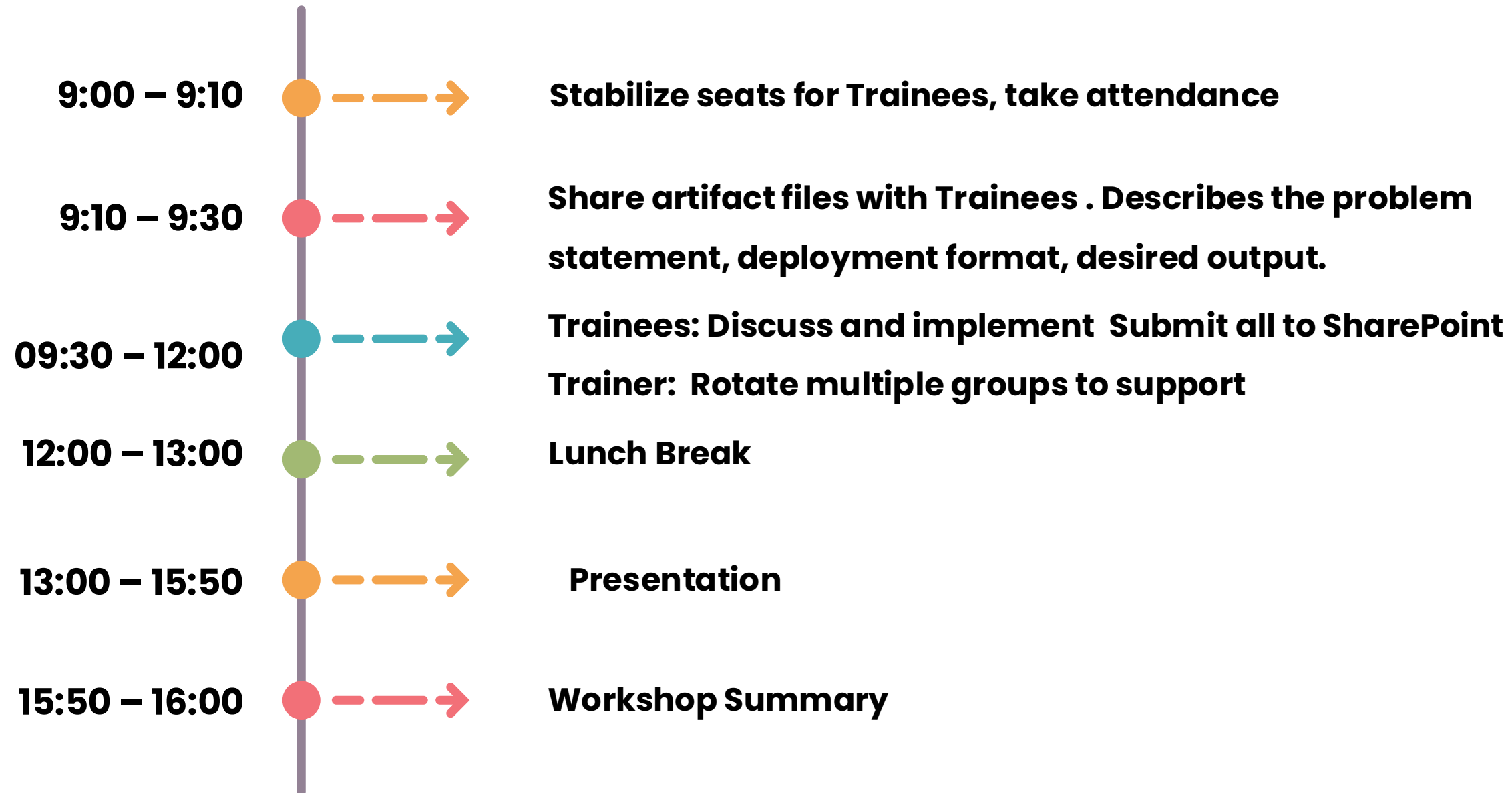
Workshop 3: Building a Chatbot System Using ChromaDB, OpenAI SDK, HuggingFace Text-to-Speech



Date



Timelines



Workshop Objective

Build a chatbot system that integrates all these technologies

What you'll learn:

 **Integrate Azure OpenAI's API with Python code**

 **Automate text generation, summarization, or other processes**

 **Build useful assistants for various tasks**

 **Handle prompts effectively**

Application Options

Choose One or Define Your Own:



Customer Support

**Responds to queries and speaks answers
aloud**



Educational Tutor

**Answers student questions with audible
explanations**



Entertainment Recommender

**Suggests movies/shows with spoken
recommendations**



Language Learning Assistant

**Helps learning with responses in target
language**

AI Tools Usage Guidelines

 Tool/Technology	 Usage
ChromaDB	Persistent storage of knowledge using vector embeddings
OpenAI SDK	Language model for conversational responses and query processing
HuggingFace TTS (VITS model)	Converts text responses into speech using a pretrained model

Teamwork Approach

-  **Teams of 4-5 participants** collaboratively develop all parts
-  **No predefined roles** - everyone contributes to all aspects
-  **Focus:** Combining data storage, language generation, and speech synthesis
-  **Development:** Shared Jupyter Notebook or minimal web-based UI



Deliverables by End of Workshop

 **ChromaDB Implementation** - storing mock data (user queries, FAQs, knowledge base)

 **HuggingFace TTS** - converting responses into spoken audio

 **Working codebase** with a simple interface (CLI or web)

 **Tested conversation logs** demonstrating multi-turn interaction

 **Team presentation** with demo, insights, and lessons learned

THANK YOU!



Ready to Build Amazing AI Tools?

Let's turn your repetitive tasks into automated solutions!